

ABET Self-Study Evaluation and Continuous Improvement Report

Course: CHE 311 – Transport Phenomena I, Fall 2024

1. Self-Study Evaluation (Findings and Interpretation)

The assessment of Student Outcome 1 (SO1) in CHE 311 was conducted using the final exam as the primary direct assessment instrument, with a total exam score of 100 points.

Overall student performance indicates **moderate attainment with significant variability across the class**.

The **average exam score of 61.50** with a **standard deviation of 18.86** reflects a broad distribution of student achievement. While a portion of the class demonstrated adequate understanding of momentum transport and fluid mechanics principles, a substantial number of students struggled with foundational and applied concepts.

The **performance indicator average of 2.35 (on a 3-point rubric scale)** exceeds the minimum attainment benchmark of 2.0, indicating that **SO1 is satisfactorily attained** for this course. However, the relatively high standard deviation of **0.81** suggests inconsistent mastery of learning outcomes across the student population.

The distribution of performance levels further highlights this variability:

- **Strong (S): 11 students**
- **Developing (D): 5 students**
- **Unsatisfactory (U): 4 students**

Although **80% of students achieved at least a developing level**, the presence of **20% unsatisfactory performance** indicates that a meaningful subset of students is not achieving the expected competency in engineering problem-solving.

These results suggest that while many students are able to apply fluid mechanics principles effectively, others struggle particularly with:

- Multi-step momentum balance problems
- Translating physical systems into governing equations
- Applying assumptions and boundary conditions appropriately
- Integrating mathematical and physical reasoning in problem-solving

The observed variability indicates differences in student preparedness and suggests that instructional support is not uniformly effective across all learners.

2. Continuous Improvement Actions

To address the identified gaps and improve consistency in student achievement, the following continuous improvement actions will be implemented:

A. Strengthening Foundational Knowledge

- Reinforce key prerequisite concepts (e.g., statics, differential equations, control volume analysis) through structured review modules at the beginning and throughout the semester.
- Integrate short conceptual refreshers before introducing complex momentum transport topics.

B. Enhancing Formative Assessment Strategy

- Introduce **frequent low-stakes assessments** such as weekly quizzes and in-class problem sets.

- Provide timely feedback to allow students to correct misconceptions early in the course.
- Use formative data to identify at-risk students before major assessments.

C. Improving Problem-Solving Instruction

- Develop and consistently apply **structured problem-solving frameworks** for fluid mechanics problems.
- Incorporate guided examples emphasizing:
 - Problem definition and system selection
 - Free-body and control volume analysis
 - Governing equation formulation
 - Interpretation of results in physical terms
- Gradually increase problem complexity to build student confidence and competency.

D. Increasing Active Learning and Engagement

- Incorporate in-class collaborative problem-solving sessions and peer instruction.
- Use conceptual questioning techniques to promote deeper understanding of fluid mechanics principles.
- Encourage students to verbalize reasoning during problem-solving activities.

E. Strengthening Academic Support Mechanisms

- Provide targeted tutoring and review sessions for students identified as at risk based on formative assessment results.
- Promote increased utilization of office hours and peer-assisted study groups.
- Offer optional review workshops prior to major exams.

F. Assessment Alignment

- Conduct a detailed review of exam items to ensure alignment with course learning objectives and instructional content.
 - Perform item analysis to identify specific topics contributing to low performance.
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3. Reassessment Plan

The effectiveness of these improvement strategies will be evaluated in the next course offering by:

- Monitoring changes in **average rubric scores and exam performance distributions**
 - Reducing the proportion of **Unsatisfactory (U) performance levels**
 - Decreasing the **standard deviation** to indicate more consistent student learning outcomes
 - Conducting comparative analysis of performance on key fluid mechanics topic areas
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4. Conclusion

Student Outcome 1 is **attained in CHE 311**, but the assessment reveals **moderate overall performance with significant variability among students**. The continuous improvement plan focuses on strengthening foundational knowledge, increasing formative feedback, enhancing structured problem-solving instruction, and improving targeted student support. These actions are expected to improve consistency of learning outcomes and reduce the proportion of underperforming students in future course offerings.